My GitHub projects!

Wen-Hsin Chen

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SKILLS

Languages: Python, C, R, Java, HTML/CSS, C++, Visual C#, SQL, BASH shell script, Haskell, Prolog

Knowledge: Machine Learning, Generative AI, Artificial Intelligence, Big Data management, satellite geospatial analysis, PyTorch, scikit-learn, Front-End Web Development, Back-End Web Development (CRUD)

EXPERIENCE

UNDERGRAD & GRADUATE RESEARCH

June 2023 – Present

Dr. Shyu, Chi-Ren, University of Missouri - Columbia

- **Neurocritical Care & AI**: Developed a chatbot that can reference multiple neurocritical care guidelines to answer domain-specific queries by **employing text embeddings and Retrieval Augumented Generation (RAG)**.
- Engineered a program that can **use AI to recommend appropriate physical therapy videos to recovering patients** at home based on their neurocritical-care conditions.
- **Geospatial database & language processing**: Developing a platform to efficiently and intelligently retrieve from a geospatial Big Data database with 162 million rows **while using LLMs and automated machine learning methods to determine how geographical environments affect pulmonary health outcomes**.

PROJECTS

RECAPAI WITH CHATGPT API & ASSEMBLYAI

- 1st-place hackathon submission & 1st ChatGPT API project: Software performs **AI summarizing (recapping), sentiment analysis, keypoints extraction, tasks extraction of an audio file** and saves AI responses into a docx, pdf, or txt
- code files - https://github.com/chenwh0/ReCapAI

Artificial General Intelligence (ARC-AGI) HACKATHON

- 1st place hackathon submission & team hackathon project: ARC-AGI puzzle sets are difficult for machines to solve and required us to explore engineering **a model that has fluid artificial intelligence, logical reasoning, can learn outside of training data**
- code files - https://github.com/chenwh0/ARC-AGI-hackathon

AI HAND GESTURE DETECTION WITH RASPBERRY PI

- Created a program using Raspberry Pi and PiCamera that **executes Youtube hotkeys based on hand gesture AI-detected from TensorFlow**
- code files - https://github.com/chenwh0/HandGestureRecognitionWithOpenCVnRaspberryPi

STORE BUSINESS WEBSITE

- **Web designed** a Store Business Website for Hong Kong Market of Columbia, MO
- Added in **back-end development** including: CRUD operations, Microsoft SQL server migration, thus allowing employees to easily create new store products or edit pricing. (https://github.com/chenwh0/hkm)

EDUCATION

AA in General Studies at Moberly Area Community College

Relevant coursework: *OOP, Relational Database Systems, Web Design, Linux Admin*

MS in Computer Science at University of Missouri

Relevant coursework: *AI, Big Data Management, Cloud Computing, Neural ML, Cyber Defense, Software Security*